

Human gut derived-organoids provide model to study gluten response and effects of microbiota derived molecules in celiac disease

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Introduction

Organoids and celiac disease (CD)

Celiac disease is an autoimmune disorder triggered by gluten consumption in genetically susceptible individuals.

In this study we demonstrate through bulk RNA-seq data analysis, that organoids are valid models to study CD, and that new DEGs in CD patients' organoids have been discovered. Here is a description of the RNA-seq pipeline step by step.

Algorithms and Results

Adapter Trimming and Quality Control

Algorithm: Trim Galore and FastQC

- ▶ Adapter trimming performed using Trim Galore.
- ▶ Cutadapt algorithm used to remove adapter sequences from 3' ends.
- ▶ Phred score cutoff: 20. Quality check using FastQC.
- ▶ FastQC provides modular analysis of data, including base sequence content, GC content, etc.

Trim Galore algorithm

Trim Galore works as follows:

1. Subtract the quality threshold from all quality scores in the read.
2. Calculate partial sums from each position to the end of the sequence.
3. Find the position where this sum reaches its minimum value.
4. Trim the read from this position to the end.

This algorithm is designed to remove low-quality bases from the 3' end of reads while allowing some higher-quality bases to remain among lower-quality ones.

QC Findings

- ▶ After QC, 0.3-0.5% of sequences were discarded.

Algorithm: SortMeRNA

- ▶ SortMeRNA was used to filter out rRNA sequences using a seeding strategy with burst tries and lookup tables.
- ▶ The core of SortMeRNA is based on an approximate seed matching strategy. This approach focuses on finding many short regions of similarity between an rRNA database and the input reads.

Workflow

1. **Input:** SortMeRNA takes as input a file of reads (FASTA or FASTQ format) and one or multiple rRNA database files.
2. **Seed Matching:** The algorithm searches for short regions of similarity (seeds) between the input reads and the rRNA database.
3. **Alignment:** For reads with matching seeds, SortMeRNA performs local alignment to confirm the match.
4. **Sorting:** The algorithm sorts the reads into two categories - those aligning to rRNA sequences and those that do not.
5. **Output:** SortMeRNA produces two output files - one containing rRNA reads and another with non-rRNA reads.

rRNA Removal Findings

- ▶ 99.97% of reads failed the threshold, meaning they did not contain rRNA sequences.

Genome Indexing and Mapping

Algorithm: STAR

- ▶ STAR software used for genome indexing (hg19) and read mapping.
- ▶ Utilizes the Maximal Mappable Prefix (MMP) method and uncompressed suffix arrays with binary search.
- ▶ Reads were 50 bp, mapped on reverse strand.

Genome Mapping Results

- ▶ 16–21 million aligned reads per sample.
- ▶ Reverse-stranded reads, typical for most library preparation protocols.

Read Quantification

Algorithm: HTSeq

- ▶ HTSeq used for read quantification.
- ▶ Genes with baseMean ≥ 10 and adjusted p-value < 0.05 were selected as differentially expressed.

Algorithm: PyDeseq (Differential expression analysis)

PyDESeq2 models raw RNA-seq count data using a negative binomial distribution. This distribution is chosen because it accounts for the overdispersion commonly observed in RNA-seq data. Dispersion Estimation: The algorithm estimates dispersion parameters independently for each gene by fitting a negative binomial generalized linear model (GLM). Normalization: PyDESeq2 normalizes the RNA-seq count data to account for differences in sequencing depth and RNA composition across samples. Model Fitting: A generalized linear model is fitted to the normalized count data using the negative binomial distribution.

Differential Expression Analysis Findings

- ▶ 83 differentially expressed genes found.
 - ▶ 45 upregulated (e.g., NLRP6, DEFA6).
 - ▶ 38 downregulated (e.g., DKK1, TFF1, TFF2).

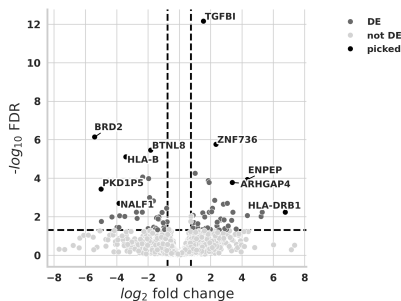


Figure 1: Volcano plot showing DEGs with high significance and/or high $\log_2$ FC

Gene Set Enrichment Analysis (GSEA)

Algorithm: GSEAPy

- ▶ GSEAPy used to compare transcriptome expression values.
- ▶ GSEA performed against 'GO Biological Process 2021' gene sets.

GSEA Findings

- ▶ “Cell-substrate junction assembly” is more associated with control samples than disease samples.
- ▶ It can be interpreted as impaired gut permeability in the disease group

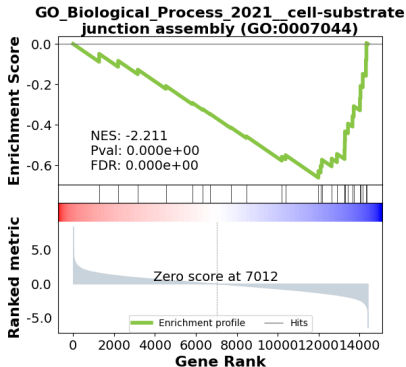


Figure 2: GSEA plot

Overview of Findings on Celiac Disease Pathogenesis

- ▶ Overexpression of CD36 a receptor for thrombospondin-1 (TSP-1), a glycoprotein involved in platelet adhesion and an endogenous inhibitor of angiogenesis
- ▶ HLA-DQ2 or HLA-DQ8 haplotypes involved in pathogenesis.
- ▶ Intestinal permeability increased due to zonulin release.

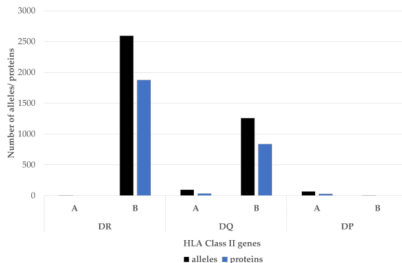


Figure 3: The HLA class II beta chains are much more polymorphic than the alpha chains. The most polymorphic HLA class II locus is HLA-DRB1-2690 alleles of this gene have been identified. Source: doi:10.3390/cells9051127

Molecular Mechanisms Involved in CD

- ▶ Enzyme TG2 plays a pivotal role in deamidation and crosslinking of gliadin peptides.
- ▶ T-cell response to gliadin contributes to autoimmunity.

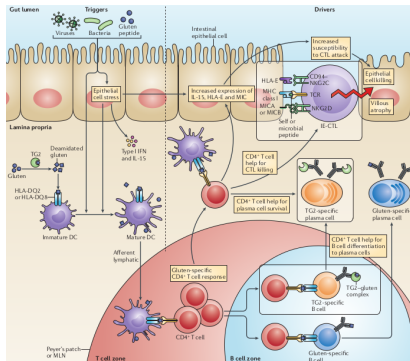


Figure 4: Autoimmune response in CD. Source: doi:10.1038/nri3407

Role of Th17 in Celiac Disease

- ▶ Overexpression of TGF β 1 supports the role of TGF- β in Th17 differentiation.
- ▶ Th17 cells drive the immune response in CD.

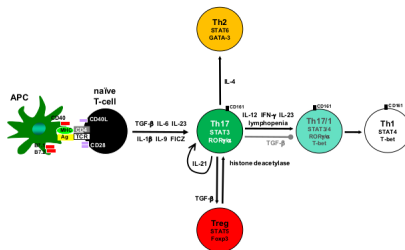
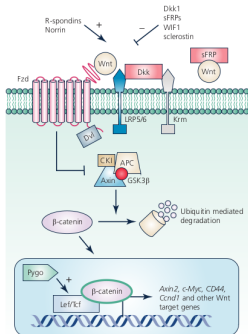


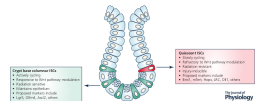
Figure 5: Molecular mechanisms of Th17 differentiation. Source: DOI: 10.1189/jlb.0211099

Wnt Signaling and Inflammation

- ▶ Downregulation of DKK1, a WNT negative regulator, correlates with increased inflammation.
- ▶ WNT signaling linked to epithelial proliferation in CD.



(a) Overview of wnt of the canonical Wnt pathway and points of regulation



(b) The intestinal epithelium contains two functionally distinct pools of intestinal stem cells (ISCs)

Figure 6: An overview of wnt signaling and ISCs. Source: DOI: 10.1113/JP271754

Overexpression of Wnt due to DKK1 underexpression

Overexpression can lead to excessive cell division in the intestinal crypts. High levels of Wnt signaling can interfere with normal cell differentiation processes, potentially leading to an accumulation of undifferentiated or partially differentiated cells. Overactive Wnt signaling can expand the stem cell compartment and disrupt the balance between self-renewal and differentiation. Wnt creates a gradient along the crypt-villus axis, influencing cell positioning and migration. Overexpression could disrupt this gradient and alter cellular organization

SMAD Pathway in CD

- ▶ Underexpression of PTPRK and involvement of the SMAD pathway in epithelial proliferation.
- ▶ Dysregulation of EGFR signaling in CD.

PTPRK is generally underexpressed in CD patients, but is usually escorted by an overexpression of TGF- β (which stimulates PTPRK expression through SMAD pathway) which seems to be a compensation mechanism.

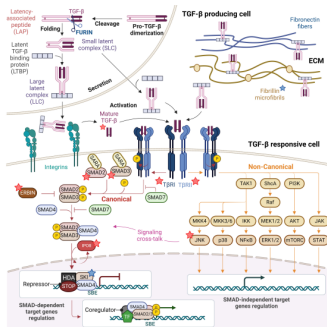


Figure 7: Role of TGFB1 in SMAD signaling. Source: doi: 10.3389/fimmu.2022.1066375

Role of TGFB1 in Containing CD Inflammation

- ▶ TGFB1 downregulation leads to immune abnormalities.
- ▶ SMAD2/3 signaling helps contain inflammation.

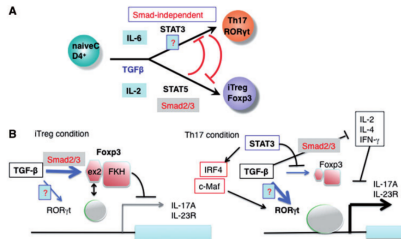


Figure 8: TGFB1 connection with SMAD pathway. Source: doi:10.1093/jb/mvq043

Role of TGFB1 in Containing CD Inflammation

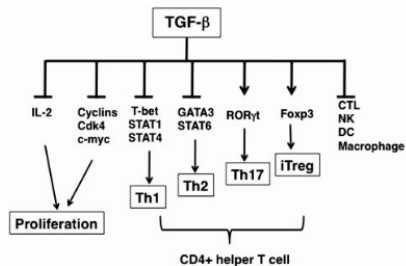


Figure 9: TGFB1 role in the immune response. Source: doi:10.1093/jb/mvq043

Key Results and Implications for CD

- ▶ NLRP6 and inflammasome involvement in CD.
- ▶ DEFA6 overexpression points to bacterial dysbiosis.
- ▶ CD36 overexpression linked to reduced angiogenesis and nutrient absorption.

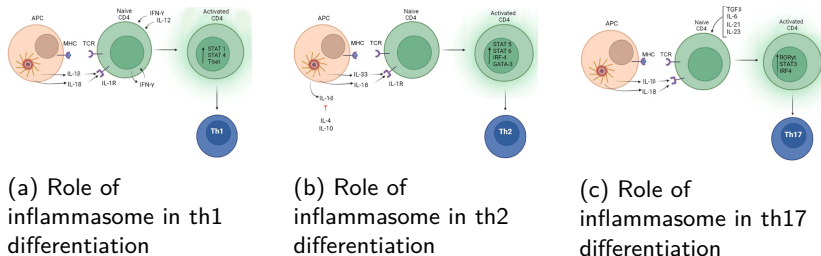
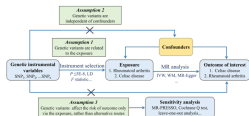


Figure 10: Different roles of inflammasome in Th differentiation. Source: doi: 10.3389/fmicb.2022.851835

Celiac disease and Rheumatoid arthritis connection



(a) A mendelian randomization study design, which proved that the prevalence of CD in RA patients is triple that of the healthy population.

HLA-DRB1 Allele	Allelic Frequency	PI	Protein Change	Performance for Celiac Disease Screening	Association Specificity	Blinding Ability of Selected RA Symptom	Study Size	Study Type	Reference
*01	0.05	positive	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,11]
*02	0.05	negative	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,12]
*03	0.05	positive	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,13]
*04	0.05	positive	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,14]
*05	0.05	positive	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,15]
*06	0.05	positive	Protein change	Protein change	Protein change	Protein change	Protein change	Protein change	[10,16]

(b) A table demonstrating the risk alleles of HLA-DRB1 for RA.

Figure 11: Connections between RA and CD through HLA-DRB1. A report confirmed that 70% of CD patients confirmed by biopsy who had HLA-Q2 also have association with HLA-DRB1*03. Sources: doi: 10.3389/fgene.2022.976579 ; doi:10.3390/cells9051127 ; DOI: 10.1097/MEG.0b013e32834d09d4.

Conclusion and Future Directions

- ▶ Implications of differential gene expression in CD progression.
- ▶ Additional studies on RA-CD connection
- ▶ Future therapies may focus on TG2 inhibitors (e.g., ZED1227).

Thank you for the attention